1.9 TEST OF HYPOTHESES FOR POPULATION MEAN
(Population variance is known)

Procedure:

Step 1 : Let μ and σ^2 be respectively the mean and the variance of the population under study, where σ^2 is known. If μ_0 is an admissible value of μ, then frame the null hypothesis as H_0: $\mu = \mu_0$ and choose the suitable alternative hypothesis from

(i) H_1: $\mu \neq \mu_0$ (ii) H_1: $\mu > \mu_0$ (iii) H_1: $\mu < \mu_0$

Step 2 : Let $(X_1, X_2, \ldots, X_n)$ be a random sample of n observations drawn from the population, where n is large ($n \geq 30$).

Step 3 : Let the level of significance be α.

Step 4 : Consider the test statistic $Z = \dfrac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$ under H_0. Here, $\bar{X}$ represents the sample mean, which is defined in Note 2. The approximate sampling distribution of the test statistic under H_0 is the $N(0,1)$ distribution.

Step 5 : Calculate the value of Z for the given sample $(x_1, x_2, \ldots, x_n)$ as

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Step 6 : Find the critical value, z_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Critical Value (z_e)	$z_{\alpha/2}$	z_α	$-z_\alpha$

Step 7 : Decide on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Rejection Rule	$\lvert z_0 \rvert \geq z_{\alpha/2}$	$z_0 > z_\alpha$	$z_0 < -z_\alpha$

Example 1.7

A company producing LED bulbs finds that mean life span of the population of its bulbs is 2000 hours with a standard derivation of 150 hours. A sample of 100 bulbs randomly chosen is found to have the mean life span of 1950 hours. Test, at 5% level of significance, whether the mean life span of the bulbs is significantly different from 2000 hours.

12th_Statistics_EM_Unit_1.indd 13 1/28/2020 4:47:33 PM